

LANGLADE COUNTY AP, WI, USA

WMO: 726626

Lat: 45.154N

Lon: 89.111W

Elev: 464

StdP: 95.87

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 04864

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.2	-23.0	-29.9	0.2	-24.8	-27.3	0.3	-22.8	11.0	-8.0	10.0	-8.7	2.4	280	0.611

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	29.0	21.9	27.6	20.8	26.3	19.7	23.6	27.0	22.4	25.9	21.3	24.7	4.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.5	18.2	25.5	21.3	16.9	24.4	20.1	15.7	23.3	72.8	27.4	68.2	25.9	64.1	24.5	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.5	8.9	8.0	-30.4	31.9	2.5	1.9	-32.2	33.3	-33.7	34.4	-35.1	35.5	-36.9	36.9		
			-30.6	25.1	2.5	1.6	-32.3	26.3	-33.8	27.2	-35.2	28.1	-36.9	29.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.4	-10.2	-9.1	-2.7	5.1	11.8	16.9	19.2	18.1	14.3	7.3	0.2	-7.3	(6)
(7)		DBStd	11.70	6.77	6.45	6.73	5.29	4.69	3.65	3.15	3.07	4.48	4.73	5.46	6.50	(7)
(8)		HDD10.0	2714	627	535	398	162	34	1	0	0	12	114	294	537	(8)
(9)		HDD18.3	4871	886	768	653	398	211	69	28	42	134	344	543	795	(9)
(10)		CDD10.0	1023	0	0	3	14	89	208	286	252	141	29	2	0	(10)
(11)		CDD18.3	136	0	0	0	0	6	26	54	35	14	1	0	0	(11)
(12)		CDH23.3	1263	0	0	0	7	83	223	504	311	124	10	0	0	(12)
(13)		CDH26.7	250	0	0	0	1	15	39	116	57	23	0	0	0	(13)
(14)	Wind	WSAvg	3.6	3.9	3.9	3.8	4.2	3.9	3.3	2.7	2.6	3.2	3.7	3.8	3.8	(14)
(15)	Precipitation	PrecAvg	786	25	23	39	67	93	105	94	91	98	73	42	34	(15)
(16)		PrecMax	979	68	51	73	123	139	190	244	151	226	142	137	66	(16)
(17)		PrecMin	644	6	8	5	26	38	44	23	25	15	11	6	14	(17)
(18)		PrecStd	102	16	11	20	30	32	40	50	31	53	37	28	16	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.8	7.5	19.0	24.0	28.7	29.9	32.2	30.7	29.0	24.9	16.4	9.5	(19)
(20)			MCWB	2.2	5.2	14.2	14.8	20.1	23.1	23.7	22.8	21.1	17.3	12.0	8.9	(20)
(21)		2%	DB	2.3	4.0	12.9	20.3	26.0	27.6	29.1	27.9	27.0	21.1	12.8	4.2	(21)
(22)			MCWB	1.2	2.3	8.6	12.4	17.8	20.8	22.4	21.1	19.8	16.3	9.7	2.8	(22)
(23)		5%	DB	0.9	2.2	9.0	17.3	22.9	26.1	27.7	27.0	24.1	17.8	10.5	2.4	(23)
(24)			MCWB	0.2	1.1	6.1	10.2	15.7	19.6	21.3	20.3	18.4	13.6	7.8	1.6	(24)
(25)		10%	DB	-0.9	0.1	6.1	14.0	20.5	24.0	26.3	25.0	22.3	15.1	7.7	1.0	(25)
(26)			MCWB	-1.3	-0.7	3.5	8.4	14.7	18.5	20.1	19.0	17.5	12.0	5.7	0.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.6	5.9	14.4	17.0	21.6	24.4	26.1	24.5	22.9	20.0	13.4	9.0	(27)
(28)			MCDWB	4.1	7.1	18.1	21.8	26.5	27.7	29.3	27.8	26.6	21.9	15.5	9.2	(28)
(29)		2%	WB	1.2	2.6	10.1	13.6	19.1	22.7	23.9	22.8	21.3	16.7	10.6	3.6	(29)
(30)			MCDWB	2.0	3.9	12.3	18.5	23.6	25.9	27.0	26.3	24.7	19.2	12.7	4.1	(30)
(31)		5%	WB	0.1	1.0	6.3	11.2	17.6	21.2	22.6	21.5	20.0	14.3	7.8	1.7	(31)
(32)			MCDWB	0.6	2.0	8.7	15.8	21.0	24.4	26.3	24.8	22.9	17.4	9.4	2.4	(32)
(33)		10%	WB	-1.3	-0.8	3.7	9.0	15.7	19.6	21.4	20.4	18.6	12.3	6.0	0.2	(33)
(34)			MCDWB	-0.8	-0.1	5.7	13.7	19.2	22.9	24.9	23.6	21.1	14.7	7.8	0.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.6	10.3	10.5	11.0	11.9	11.3	12.1	11.9	11.8	9.7	7.8	7.7	(35)
(36)			MCDBR	8.3	10.1	12.3	16.6	15.2	13.2	13.4	13.6	13.9	13.8	11.8	8.1	(36)
(37)			MCWBR	7.3	8.1	8.4	9.7	8.2	7.3	7.2	7.1	7.6	8.6	8.5	6.9	(37)
(38)		5% WB	MCDBR	7.9	9.2	11.3	14.7	12.3	11.4	11.5	11.4	11.8	10.8	7.5	(38)	
(39)			MCWBR	7.1	7.7	8.3	9.7	7.6	7.3	7.2	6.9	7.2	8.4	8.5	6.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.242	0.279	0.314	0.338	0.376	0.375	0.367	0.369	0.346	0.309	0.282	0.252	(40)	
(41)		taud	2.397	2.283	2.355	2.402	2.331	2.358	2.397	2.390	2.456	2.556	2.535	2.450	(41)	
(42)		Ebn at Noon	896	906	920	923	892	892	894	881	876	872	841	846	(42)	
(43)		Edh at Noon	71	98	106	111	123	120	114	111	96	75	64	61	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.59	2.70	3.88	4.57	5.41	5.87	6.05	5.14	4.04	2.52	1.54	1.14	(44)	
(45)		RadStd	0.17	0.27	0.40	0.45	0.44	0.46	0.32	0.34	0.28	0.26	0.16	0.12	(45)	

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page